

Pigeonhole Principle

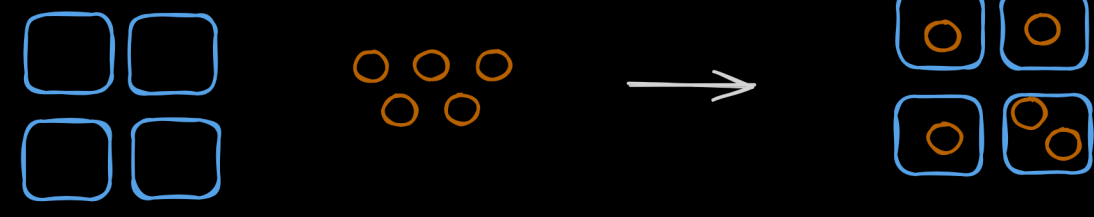
If you have more pigeons than pigeonholes, and you try to stuff the pigeons into the holes, then at least one hole must contain at least two pigeons.

The idea that having more pigeons than pigeonholes requires a pigeonhole with more than one pigeon is seemingly trivial, but it turns out to be important enough that it has a name:

A few examples will convince you of its power.

Pigeonhole Principle (naive form):

If more than n objects are placed into n boxes, then at least one box must contain more than one object.



Example 1:

Every point on the plane is colored either red or blue. Prove that no matter how the coloring is done, there must exist two points, exactly a mile apart, that are the same color.

Solution:

Just messing around quickly yields a solution. Pick a point, any point. Without loss of generality it is red. Draw a circle of radius one mile with this point as center. If any point on the circumference of this circle is red, we are done. If all the points are blue, we are still done, for we can find two points on the circumference that are one mile apart.

That wasn't hard, but that wasn't the pigeonhole solution. Consider this: just imagine the vertices of an equilateral triangle with side length one mile. There are three vertices, but only two colors available. The pigeonhole principle tells us that two vertices must be the same color!

Example 2:

A box contains three pairs of socks colored red, blue, and white, respectively. Suppose I take out the socks without looking at them. How many socks must I take out to be sure that they will include a matching pair?

Solution

If I take only 2 or 3 socks, it is possible that they are all different.

For example, they may be one red and one blue; or one red, one blue, and one white. But if I take out 4 socks, these must include a matching pair.

Here the 4 chosen socks are the "objects" and the 3 colors are the "boxes"; by PHP, it follows that at least two of the four chosen must have the same color and hence must be a matching pair. Thus the minimum number of socks to be taken out is 4.

Quick and beautiful solutions are characteristic of pigeonhole problems. The above example was quite simple. Solving most pigeonhole problems is often a threepart process:

1. Recognize that the problem might require the pigeonhole principle.
2. Figure out what the pigeons will be and what the holes must be. This is frequently the crux move.
3. After applying the pigeonhole principle, there is often more work to be done. Sometimes the pigeonhole principle yields only the "penultimate step," and sometimes just an intermediate result. The skilled problem solver plans for this when thinking of a strategy.

Example 3:

Show that given a set of n positive integers, there exists a non-empty subset whose sum is divisible by n .

Solution:

Let the n integers be denoted by $a_1, a_2, \dots, a_n$. Form the n sums

$$S_1 = a_1$$

$$S_2 = a_1 + a_2$$

⋮

$$S_i = a_1 + a_2 + \dots + a_i$$

⋮

$$S_n = a_1 + a_2 + a_3 + \dots + a_n$$

If one of these sums is divisible by n , then we are done.

Otherwise, by the pigeonhole principle, at least two of the sums must have the same remainder when divided by n (since only $n-1$ distinct remainders are possible).

Pick two such sums S_i and S_j with $j > i$.

Then it follows that

$$S_j - S_i = a_{[i+1]} + \dots + a_{[j]}$$

must be divisible by n .

Note: If a and b have the same remainder when divided by n , then $a - b$ is divisible by n .

Example 4:

(IMO 1972) Prove that from a set of ten distinct two-digit numbers (in the decimal system), it is possible to select two disjoint subsets whose members have the same sum.

Solution:

We want to have two subsets have the same sum, so it is reasonable to make the subsets be the pigeons, and the sums be the holes.

How do we do this precisely?

First, let's look at the sums. The smallest possible sum is 10 and the largest possible sum is $99 + 98 + 97 + \dots + 90$.

Consequently, there are $945 - 10 + 1 = 936$ different sums.

Now we need to count the number of pigeons. The number of subsets is just $2^{10} = 1024$.

Since $1024 > 936$, there are more pigeons than holes, so we are done: two of the subsets must have the same sum.

Pigeonhole Principle Form 1

If you have p pigeons and h holes, then at least one of the holes contains at least $\lceil p/h \rceil$ pigeons

Generalized Version

If more than $k \cdot n$ objects are placed into n boxes, then at least one box contains more than k objects.